

UNIT 6

FLUID STATICS



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FLUID STATICS:

"The study of fluid behavior in rest and its applications is called fluid statics."

PRESSURE:

"Force per unit area is called pressure."

$$P = \frac{F}{A}$$

UNIT:

S.I. unit of pressure is Pascal (Pa) or N/m².

PRESSURE IN LIQUID:

The pressure exerted at any point on area 'A' is given by:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Force} = \text{Weight}$$

$$\text{Pressure} = \frac{W}{A}$$

$$W = \rho \cdot A \cdot h \cdot g$$

$$\text{Pressure} = \frac{A \cdot h \cdot \rho \cdot g}{A}$$

$$P = \rho \cdot h \cdot g$$

PASCAL'S PRINCIPLE:

"When an external pressure applied is applied on a stationary liquid then the liquid transmits undiminished equally in all direction and acts perpendicular to the walls of the container."

APPLICATIONS OF PASCAL'S PRINCIPLE:

- Hydraulic brake system.
- Hydraulic lift OR Hydraulic Press.

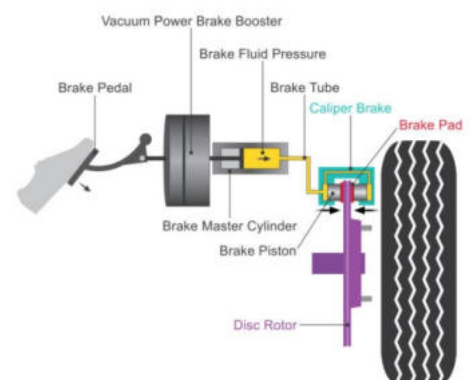
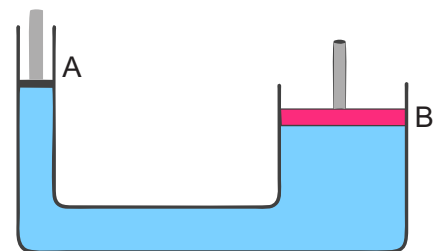
HYDRAULIC BRAKE SYSTEM:

CONSTRUCTION:

It consists of a master cylinder provided with an oil tight piston which is connected to the brake paddle. Each wheel of the car is provided with a smaller cylinder with an oil tight piston in it. These cylinders are connected through tubes to the master cylinder. Suitable oil (Brake oil) filled in this tube.

WORKING:

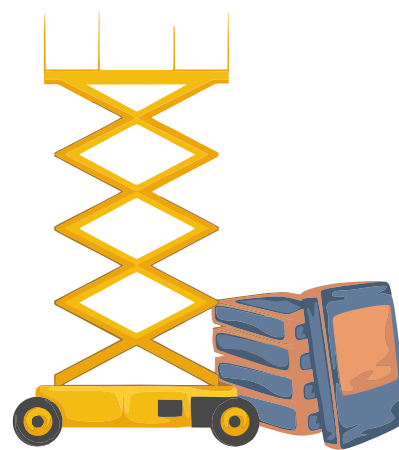
When brake paddle is pressed excess pressure in this master cylinder is transmitted equally to all the cylinders in which piston moves outward thus pressing the brake shoe against the inner surface of the wheel a large friction force is setup which tends to stop the wheel. As soon as the paddle is released, spring brings the brake shoe back to its original position.



HYDRAULIC LIFT:

In a hydraulic lift a platform is attached to an oil tight piston which moves in a wide cylinder. This cylinder is connected to a smaller cylinder which is also provided with an oil tight piston. A suitable incompressible liquid is filled in the two cylinders, and external pressure is applied on the smaller piston which is transmitted into equally, due to which the platform lifts heavy object lying on it.

$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

**ARCHIMEDES PRINCIPLE OR LAW OF FLOATATION:**

"When a body is wholly or partially immersed in a liquid, it experiences an up thrust which equal to weight of the liquid displaced."

Apparent weight = Actual weight – Upthrust

APPLICATION OF ARCHIMEDES PRINCIPLE:

- i. To find out the specific gravity.
- ii. To find out the Density of oil.
- iii. To find out the Density of cork etc.

BOUNCANCY:

"The property of a liquid by virtue of which it exerts an upward force or up thrust on a body which is wholly or partially immersed in the liquid is called buoyancy."

BOUYANT FORCE:

"The upward force on a body which is equal to the weight of the liquid displaced by the body is called buoyant force."

When an object is immersed in a liquid it is acted upon by two forces:

- i. The weight of the object.
- ii. The buoyant force acting vertically upward.

EXPRESSION FOR BOUYANT FORCE:

The pressure exerted at any point on area 'A' is given by:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Buoyant force} = \text{Pressure} \times \text{Area}$$

$$F_U = P \times A$$

$$\text{Where } P = \rho gh$$

$$F_U = \rho gh \times A$$

$$F_U = \rho gV$$

LAW OF FLOATATION:

- If weight of a body is greater than the up thrust then the body sinks in the liquid.
- If weight of the body is less than the up thrust the body will float on the liquid.
- If weight of a body is equal to the up thrust the body will float partially immersing in the liquid.

EXAMPLES OF FLOATATION:**1. Cork:**

A cork floats on the surface of water because the up thrust of water on it's greater than its weight.

2. Ship:

A ship is also designed that its volume is very large. Hence the buoyant force which is equal to the weight of equal volume of water displaced by the ship is very large and is greater than its weight. Hence the ship floats on the surface of the water in a sea.

3. Needle:

A needle is made of iron sinks in water because the buoyant force acting on it is less than its weight. It is due to the fact the volume of a needle is very small and hence the weight of water displaced by the needle is less than the weight of the needle.

4. Submarines:**Definition:**

"Submarine can float on the surface of water and when needed they can dive into water. They are fitted with large hollow ballast tank."

5. In order to dive in water, they fill the tank with water. This increases the weight of the submarine and submerges in water. To bring the submarine on the surface extra load due to the water in the tank is removed. This is done by forcing compressed air into the tank which expels water from the tank.

SURFACE TENSION:

"Surface tension is the property of a liquid by virtue of which the free surface of a liquid behaves like a stretched-membrane tending to decrease the surface area."

If " σ " represents the surface tension then:

$$\sigma = \frac{F}{L}$$

$$\sigma = \frac{F}{2L}$$

Unit:

In S.I or M.K.S system, unit of surface tension is N/m.

EXAMPLES OF SURFACE TENSION:**1. Steel Needle:**

In general, a steel needle, if dropped in water, will sink because the density of steel is greater than that of water. However, if the needle is made slightly oily and then placed on the surface of water in a horizontal position, it will float leaving a depression of water under it. It is the surface tension of water which supports the needle and stops it from sinking.

2. Metallic Ring (Thin Film):

A metallic ring is dipped into a soap solution. A light moist loop of cotton thread of any shape is gently placed over the soap film. When the soap film within the loop is touched with a hot needle so as to break it, the loop takes a circular shape. Every point on the thread experiences radially outward force of equal magnitude due to surface tension in the soap film outside the loop. This force being due to surface tension in the plane of the soap film.

3. Drop of Olive Oil:

When a drop of olive oil is left gently inside a mixture of spirit and water in such a way that density of mixture being equal to that of olive oil with the help of pipette. The olive oil drop assumes a perfect spherical shape. Due to surface tension, a liquid tends to keep its surface area minimum. For a given volume, the surface area of the sphere is minimum. Due to the reason the rain drops are spherical in shape.

4. Water Droplets Falling from a Tap:

Water droplets falling from a tap are spherical in shape. It is due to surface tension that minimizes the surface of the drop similarly.

5. Capillary Rise or Capillarity:

It is a phenomenon in which liquid spontaneously rises or falls in a narrow space such as a thin tube or in the voids of a porous material. Surface tension is an important factor in the phenomenon of capillarity. The height of the column can be calculated by:

$$h = \frac{2T}{\rho g r}$$

SCIENTIFIC REASONS:**Q1. An object immersed in water appears lighter. Why?**

When an object is immersed in water it experiences an up thrust exerted by water in upward direction. Hence, it appears light in weight.

Q2. Why does a ship made up of iron float on water?

The ship is made in such a way that it is hollow from within. Therefore, only a fraction of volume of ship can displace water equal to its own weight. Thus, a fairly large fraction of ship stays above the level of water.

Q3. An ink pad absorbs ink. Why?

An ink pad works on the principle of capillary action. Ink pad is porous therefore when ink is put on it, the ink rises in its fine pores and gets absorbed by capillary action.

Q4. A ship can be loaded more on sea in comparison to than on river. Why?

The density of river water is less than that of sea water; therefore ship has to displace more volume of water so as to support its own weight. The sea water, however, provides greater up thrust than river water due to which it can be loaded more.

Q5. An ice cube floats on water. Why?

The density of an ice cube is lower than that of water and it has a large volume. Therefore, the ice cube displaces the volume of water equal to its own weight and hence it floats.

Q6. Ongoing to high altitudes there is a danger of bursting of blood capillaries. Why?

The atmospheric pressure decreases to a great extent on moving at high altitudes. The internal pressure of the blood capillaries is greater than the external atmospheric pressure. Therefore, there is danger of bursting of capillaries.

Q7. It is easier to lift an object in water than in air. Why?

According to Archimedes principle, when an object is immersed in water it experiences a fall in its weight. This fall in its weight is equal to the weight of the water displaced by it. Hence, it is easier to lift an object in water than in air.

Q8. It is advised to empty a pen's ink before moving on a plane. Why? Why does a fountain pen leak at higher altitudes?

On higher altitudes a fountain pen starts leaking because air inside the pen is at higher pressure than air outside as the atmospheric pressure decreases considerably on higher altitudes.

Q9. Water pipes burst in winters. Why?

The temperature of water falls in winters due to which the water inside the pipe freezes and its volume increases. The ice tends to occupy the space equal to its volume. Therefore, water pipe bursts.

Q10. Why are raindrops spherical?

The free surface of a liquid tends to occupy minimum surface area. The surface area of water decreases due to surface tension. The surface area of a sphere is minimum. Therefore, raindrops are spherical in shape.

Q11. A train stops on pulling the chain. Why?

When the chain is pulled the rod comes out of vacuum chamber and air fills its space. Due to which the brakes are applied on the wheels and the train stops.

Q12. Why is mercury filled in a barometer and not water?

Since the relative density of mercury is 13.6 times greater than that of water, only 76 cm height of mercury column is needed to balance the normal atmospheric pressure. On the other hand, with water the length of the column needed will be nearly 10 m which is inconveniently large. Apart from it, the surface of mercury is shining and opaque and thus it is easily seen while taking the observations.

Q13. Presence of the atmosphere increases the length of the day. Why?

Due to the refraction of sunlight while passing through the layers of atmosphere, the sun can be seen even when it is below the horizon at sunrise. Similarly, at sunset also can be seen when below the horizon. Therefore, length of the day increases by about 8 minutes.

Q14. Why are the walls of a dam thick at the bottom and thin at the top?

The pressure of water increases with depth. Therefore, there is a high pressure of water on the walls of the dam at the bottom. That is why the walls of the dam are thickest at the bottom and their width gradually decreases towards the top.

Q15. Water converts into steam on heating. Why?

The inter-molecular space between the molecules of a liquid is larger than solids and lesser than gases. Therefore, the force of attraction between the liquid molecules is lesser than the molecules of a solid. The water molecules are therefore free to move within the container. When water is heated to its boiling point the inter-molecular force between its molecules becomes negligible. The kinetic energy of its molecules increases to such an extent that they begin to move randomly with increased velocity in all directions and finally leave the water surface, converting into vapor.

Q16. Why does camphor dance on water?

Camphor is soluble in water but irregular in size. A part of camphor gets dissolved in water more than the other part. When camphor dissolves in water the surface tension of water decreases. The surface tension of water remains more where it does not dissolve. As a result, the surface tension of water on one side of camphor piece is less as compared to that on other side, hence a resultant force (from the direction of surface tension to high surface tension) acts on the camphor piece and it moves towards the direction of the resultant force. Again, the same process continues at the place where it reaches and hence, it appears to dance on the water surface.

Q17. Why does a human being not feel atmospheric pressure?

The average area of a human body is 16 square feet and the atmospheric pressure is about 15 tons. But we do not feel the presence of such a heavy weight. This is because the air that we inhale is filled in lungs and the lungs send this air to blood vessels at atmospheric pressure. The oxygen from air is used for purification of blood. Therefore, the same internal pressure is exerted from within the body which balances the external atmospheric pressure and we do not feel this atmospheric pressure, exerted on us externally.

Q18. When a liquid is left after stirring it comes to rest. why?

When liquid in a vessel is left after stirring it comes to rest after some time. If the liquid is not viscous it does not come to rest after being stirred. Thick liquids are more viscous (like coal tar, glycerin, honey) therefore thick fluids come to rest sooner than thinner fluids. If we drop coal tar on a table, it will soon come to rest. On the other hand, if we drop water on a table it will flow to a longer distance.

Q19. Why is oil sprayed on sea waves to calm them?

This is because on spraying oil, the breeze spreads the oil on the sea water in its own direction. The surface tension of sea water without oil is greater than that of oily water. Hence, the water without oil pulls the oily water against the direction of breeze and the sea waves become calm.

Q20. Why does a nail sink but a ship having a huge mass floats on water?

Nail is a solid body because it does not contain air whereas a ship is a large hollow body because it contains air. Density of nail is more than density of water. Anything whose density is more than that of water will sink into it. While the density of a ship is less than that of water. Hence nail sinks but a ship floats on water.

Q21. How do fish and other aquatic animals survive in extremely cold water in rivers, lakes, and seas?

When atmospheric temperature falls, the cooler and denser water (4°C) flows to bottom because the density of water at 4°C is maximum. If the temperature further falls below 4°C then the water at the surface expands (Anomalous expansion) and become lighter and does not sink. The ice continuously floats up the surface, because it is less dense than water, while water near the bottoms remains at 4°C . This helps fish and other aquatic animals to survive in extreme cold water.

Q22. Saline water exerts more pressure than pure water why?

Due to salinity water the density of the saline water is more than the pure water and hence it exerts more pressure.

Q23. Why body floats on the surface of the liquid?

Body floats on the surface of the liquid when the weight of the body is balanced by the up thrust force.

Q24. Why submarine submerges into water?

There are large hollow tanks fitted in the submarine. These tanks are filled with water increasing the weight of the submarine. Due to increase in the weight of submarine it submerges into water.

"The greatest weapon in the world is not nuclear weapon but the human mind."

Dr Abdul Qadeer Khan

